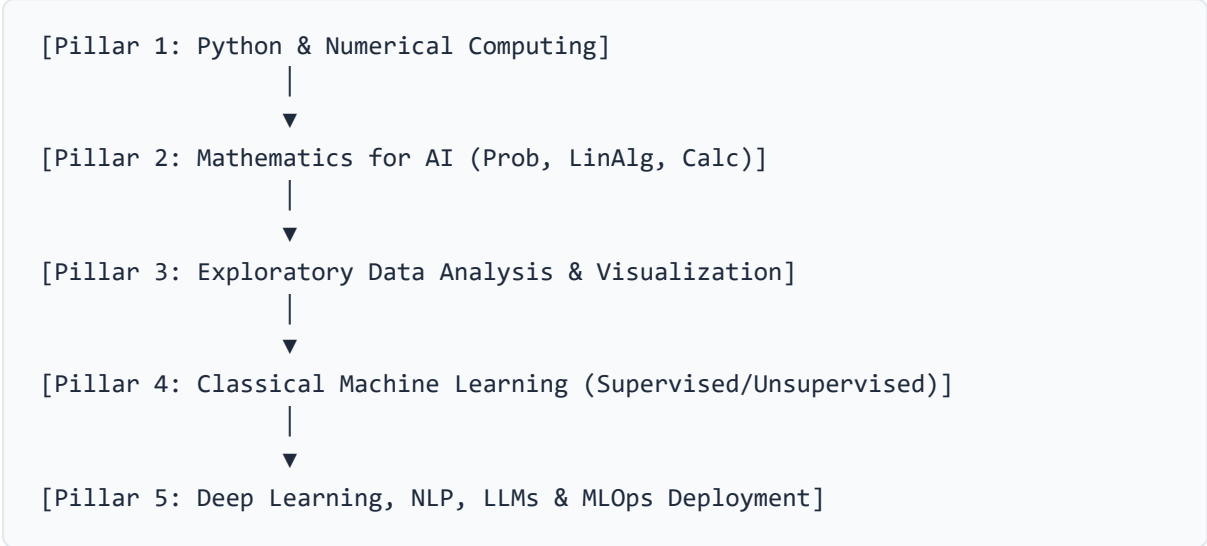
Notes 01.02: Curriculum Architecture: What Will We Learn?

1. The Five Foundational Pillars of Modern AI

Mastering Artificial Intelligence requires a structured progression across 5 interconnected technical domains:



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2. Pillar-by-Pillar Technical Scope

Pillar 1: Python Programming & Numerical Computing (Days 1–17)

- Core Python, OOP, functional patterns, memory management.
- High-performance numerical arrays with **NumPy** (vectorization, broadcasting).
- Tabular data manipulation and relational merging with **Pandas**.
- Web scraping, APIs, and data ingestion pipelines.

Pillar 2: Mathematics for AI (Days 20–22)

- **Probability Theory:** Kolmogorov axioms, Bayes' Theorem, PDF/PMF, Gaussian distributions.
- **Linear Algebra:** Vectors, matrices, eigenvalues, SVD, PCA geometric transformations.
- **Multivariable Calculus:** Partial derivatives, gradients, Jacobians, Hessians, Convex Optimization.

Pillar 3: Data Visualization & EDA (Days 18–19)

- Edward Tufte principles (Data-Ink ratio).
- **Matplotlib** 2D plotting, subplots, multi-panel figures.
- **Seaborn** statistical distributions, categorical heatmaps, pairplots.

Pillar 4: Classical Machine Learning (Days 23–35)

- Regression: Simple/Multiple Linear Regression, OLS, Ridge (L2), Lasso (L1), ElasticNet.
- Classification: Logistic Regression, Naive Bayes, k-Nearest Neighbors (kNN), Support Vector Machines (SVM), Decision Trees, Random Forests, XGBoost.
- Unsupervised: K-Means Clustering, Hierarchical Clustering, DBSCAN, PCA dimensionality reduction.

Pillar 5: Deep Learning, LLMs & MLOps (Days 36–60)

- Neural network forward propagation, backpropagation calculus, activation functions.
- Computer Vision: Convolutional Neural Networks (CNNs), transfer learning.
- NLP & Transformers: Self-Attention mechanisms, BERT, GPT, fine-tuning LLMs.
- Production MLOps: Flask, FastAPI, Docker containerization, CI/CD pipelines.